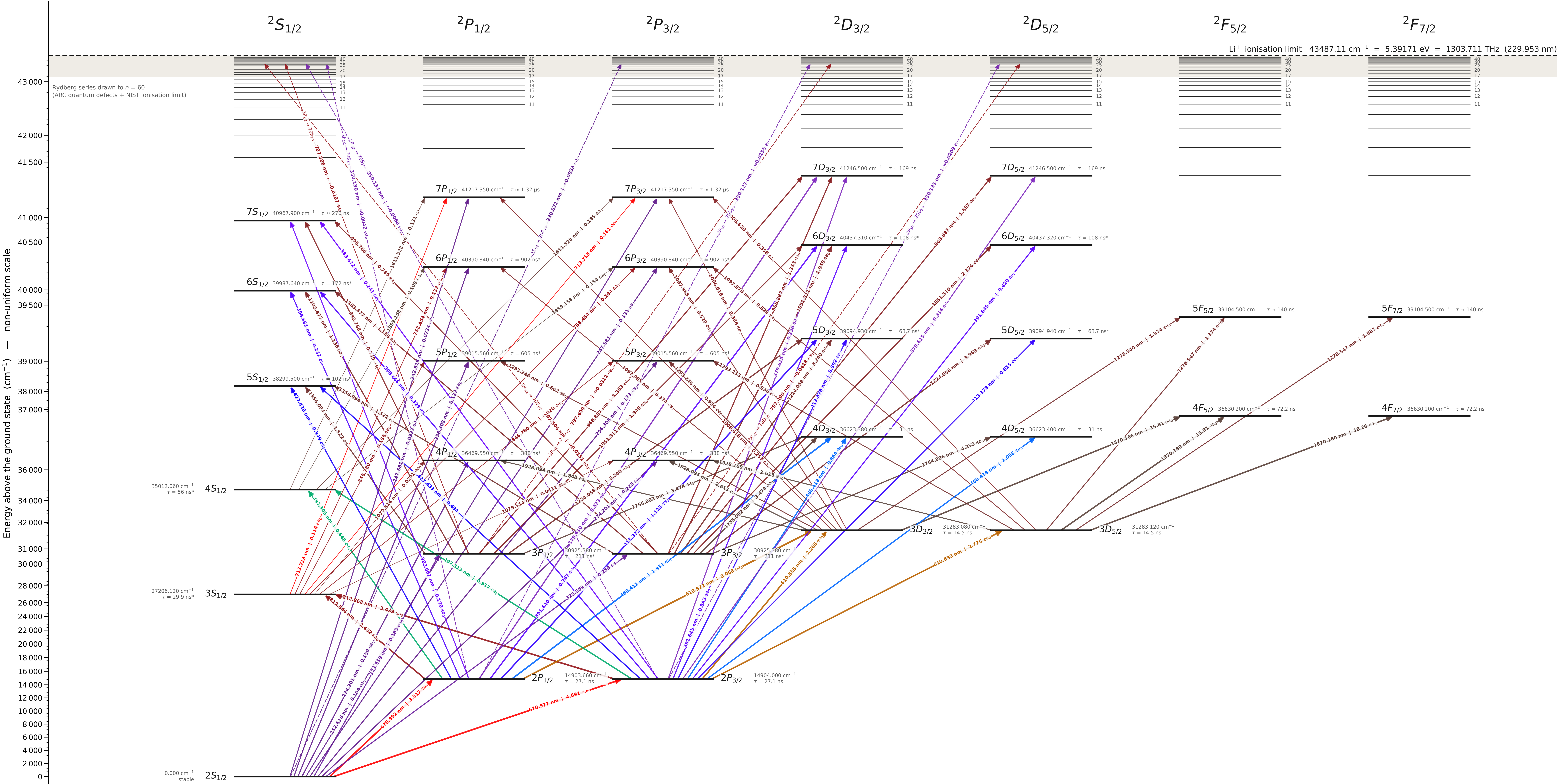
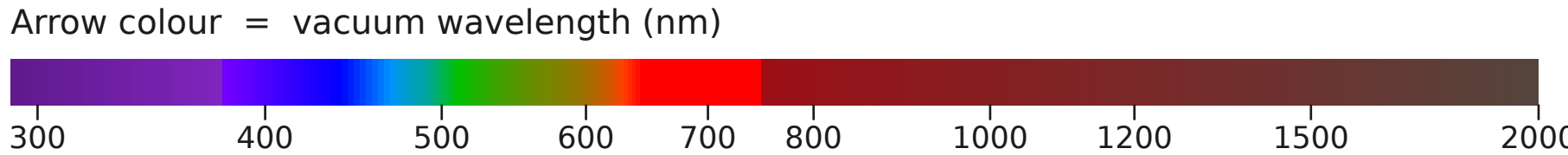


Lithium-6

⁶Li (Li II), nuclear spin *I* = 1. Energy levels and transitions below 2 μm: 32 levels, 85 lines.

Arrow colour = vacuum wavelength (nm) | reduced dipole matrix element $\langle J||er||J' \rangle$ | in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked ≈).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\tilde{\nu}$ (cm ⁻¹)	$\langle J er J' \rangle$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Typical use
2P _{1/2} – 3D _{3/2}	610.5222	610.3533	491.0427	16379.420	5.066	5.714e+07	10.980	83.3	
2S _{1/2} – 2P _{1/2}	670.9925	670.8073	446.7896	14903.297	3.317	3.690e+07	5.872	100.0	D1 line
2S _{1/2} – 2P _{3/2}	670.9774	670.7922	446.7996	14903.632	4.691	3.689e+07	5.872	100.0	D2 line
2P _{3/2} – 3S _{1/2}	812.8680	812.6446	368.8083	12302.120	3.439	2.231e+07	5.329	66.7	
2P _{1/2} – 4D _{3/2}	460.4111	460.2821	651.1408	21719.720	1.931	1.935e+07	5.134	64.5	
2P _{3/2} – 3D _{5/2}	610.5334	610.3644	491.0337	16379.120	2.775	1.143e+07	10.980	100.0	
2P _{3/2} – 3D _{3/2}	610.5349	610.3659	491.0325	16379.080	2.266	1.143e+07	10.980	16.7	
2P _{1/2} – 3S _{1/2}	812.8456	812.6221	368.8185	12302.460	2.432	1.116e+07	5.329	33.3	
2P _{1/2} – 5D _{3/2}	413.3723	413.2557	725.2360	24191.270	1.123	9.040e+06	2.498	57.5	
2P _{3/2} – 4S _{1/2}	497.3130	497.1743	602.8245	20108.060	0.917	6.919e+06	2.840	38.8	
2P _{1/2} – 6D _{3/2}	391.6401	391.5292	765.4796	25533.650	0.767	4.964e+06	1.469	53.9	
2P _{3/2} – 4D _{5/2}	460.4179	460.2889	651.1312	21719.400	1.058	3.871e+06	5.134	77.4	
2P _{3/2} – 4D _{3/2}	460.4183	460.2893	651.1306	21719.380	0.864	3.871e+06	5.134	12.9	
2P _{1/2} – 4S _{1/2}	497.3046	497.1659	602.8347	20108.400	0.648	3.460e+06	2.840	19.4	
2P _{3/2} – 5S _{1/2}	427.4326	427.3124	701.3795	23395.500	0.494	3.170e+06	1.560	32.2	
2P _{1/2} – 7D _{3/2}	379.6098	379.5020	789.7385	26342.840	0.573	3.041e+06	0.941	52.0	
2P _{3/2} – 5D _{5/2}	413.3779	413.2613	725.2261	24190.940	0.615	1.810e+06	2.498	69.0	

λ and ν from NIST level energies (centre of gravity, no hyperfine structure); λ air uses standard air (Ciddor).
Γ/2π: natural linewidth of the upper level, 1/(2πτ). Branch: share of the upper level's decays that takes this line.

Rydberg excitation wavelengths (vacuum, nm)

Lower → series	n = 30	n = 40	n = 50	n = 70	n = 100	n → ∞	d ₇₀	ARC
2S _{1/2} → nP _{3/2}	230.602	230.317	230.186	230.072	230.011	229.953	0.0033	
2P _{3/2} → nS _{1/2}	351.397	350.715	350.404	350.134	349.992	349.857	0.0060	
2P _{3/2} → nD _{5/2}	351.356	350.698	350.395	350.131	349.991	349.857	0.0209	
3S _{1/2} → nP _{3/2}	618.862	616.817	615.877	615.060	614.628	614.213	0.0052	
3P _{3/2} → nS _{1/2}	804.085	800.527	798.905	797.506	796.770	796.068	0.0151	
3P _{3/2} → nD _{5/2}	803.872	800.439	798.860	797.490	796.764	796.068	0.0418	
3D _{5/2} → nP _{3/2}	827.699	824.045	822.367	820.912	820.142	819.404	0.0077	
3D _{5/2} → nF _{7/2}	827.672	824.034	822.361	820.910	820.141	819.404	0.0449	
4S _{1/2} → nP _{3/2}	1197.210	1189.582	1186.088	1183.064	1181.465	1179.933	0.0070	
4P _{3/2} → nS _{1/2}	1450.888	1439.347	1434.111	1429.610	1427.245	1424.996	0.0276	
4P _{3/2} → nD _{5/2}	1450.194	1439.061	1433.965	1429.558	1427.228	1424.996	0.0670	
4D _{5/2} → nP _{3/2}	1483.369	1471.676	1466.331	1461.712	1459.272	1456.937	0.0184	
4D _{5/2} → nF _{7/2}	1483.284	1471.641	1466.314	1461.706	1459.270	1456.937	0.0837	
4F _{7/2} → nD _{5/2}	1484.786	1473.117	1467.778	1463.161	1460.720	1458.382	0.0080	
5S _{1/2} → nP _{3/2}	1974.212	1953.555	1944.148	1936.037	1931.758	1927.668	0.0086	

Rydberg term energies from ARC quantum defects + NIST ionisation limit, lower level from NIST. d₇₀ = $\langle J||er||J' \rangle$ to *n* = 70 in ea₀ (calculated).
S → nP values come from a near-cancelling integral; treat them as an order of magnitude.

⁶Li hyperfine structure (*I* = 1)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2S _{1/2}	152.136841	–	F=1/2: -152.14 F=3/2: +76.07
2P _{1/2}	17.4017(9)	–	F=1/2: -17.40 F=3/2: +8.70
2P _{3/2}	-1.155(8)	-0.10(14)	F=1/2: +2.76 F=3/2: +1.25 F=5/2: -1.76
3S _{1/2}	35.274(7)	–	F=1/2: -35.27 F=3/2: +17.64
3P _{1/2}	5.3(4)	–	F=1/2: -5.30 F=3/2: +2.65
3P _{3/2}	-0.400(20)	–	F=1/2: +1.00 F=3/2: +0.40 F=5/2: -0.60
4S _{1/2}	13.5(7)	–	F=1/2: -13.50 F=3/2: +6.75

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).